Feature Engineering

Feature engineering

- A feature is an input variable used by a machine learning algorithm to make predictions.
  - For example, when predicting house prices, features may include the area of the house, number of bedrooms, location, and age of the building.
- Feature engineering is the process of creating, transforming, selecting, or extracting features from raw data to improve the performance of a machine learning model.
- **Why do we need feature engineering?**
  - The original data may not directly express all the useful relationships. Feature engineering helps represent these relationships in a form that a model can learn.
  - The main goals are to:
    - Improve prediction accuracy.
    - Represent useful patterns in the data.
    - Reduce irrelevant information.
    - Make data suitable for machine learning algorithms.

- **Major techniques of feature engineering**

- **1. Feature transformation**
    - Transform existing features into a more useful representation.
        - Example: If income values are highly skewed, we can use a logarithmic transformation.
- **2. Encoding categorical features**
    - Convert categorical values into numerical representations.
        - Example: A Color feature contains Red, Blue, and Green.
        - One-hot encoding may be used to represent these categories as separate binary columns.
- **3. Feature creation**
    - Create new features from existing variables using domain knowledge or mathematical relationships.
        - Example: If we know a person's date of birth, we can calculate their age.
        - Another example is creating BMI from height and weight
- **4. Feature selection**
    - Select the most relevant features and remove irrelevant or redundant ones.
        - Example: For predicting house prices, house area and location may be useful, whereas a randomly assigned record ID usually is not.
        - Common techniques include correlation analysis, mutual information, and recursive feature elimination.

- 5. Feature extraction
  - Transform existing data into a new representation that captures useful information.
    - Example: Principal Component Analysis (PCA) transforms correlated numerical features into a smaller number of components.
    - In image processing, features can also be extracted using methods such as HOG or deep neural networks.
- 6. Feature scaling
  - Bring numerical features to comparable scales.
  - Scaling is especially useful for algorithms such as KNN, SVM, and many gradient-based models.

4. Feature engineering vs. feature selection vs. feature extraction

These terms are related but different.

Concept	Main purpose	Example
Feature engineering	Broad process of preparing and improving features	Create price per square foot
Feature selection	Choose a subset of existing features	Retain area and location
Feature extraction	Derive a new representation from existing data	PCA components
Feature transformation	Change the representation of a feature	Log-transform income

Feature engineering is the broader concept and can include all these operations.

- **Is feature engineering always necessary?**
  - No. Its importance depends on the data and the algorithm.
  - Traditional ML: Decision Trees, SVM, KNN, and Logistic Regression can benefit substantially from carefully engineered features.
  - Deep learning: Neural networks can automatically learn useful representations from raw images, audio, and text. However, preprocessing and domain-informed feature creation can still be valuable.
  - Time-series forecasting: Features such as previous month's sales, moving averages, and day of the week can be useful.
  - Medical data: Features such as BMI, laboratory ratios, and clinically meaningful combinations may improve prediction, provided they are valid and available at prediction time.